

Submission

Objective

You are provided with a reference project using HDFS, Hive, PySpark, and Medallion Architecture. You are allowed to share and reuse common utility code, but you must prove execution by submitting Hive and HDFS evidence.

Mandatory Evidence (Screenshots Required)

Each student **MUST** submit screenshots of the following:

Hive Database & Table Creation

Screenshot must show:

```
SHOW DATABASES;  
USE iphone_analytics;  
SHOW TABLES;
```

Evidence must include:

- **bronze_*** tables
- **silver_*** tables
- **gold_*** / **fact_*** / **dim_*** tables

✗ No screenshot → No marks.

```
hive (default)> SHOW DATABASES;
OK
default
hive_test
iphone_analytics
xyz
Time taken: 0.427 seconds, Fetched: 4 row(s)
hive (default)> USE iphone_analytics;
OK
Time taken: 0.069 seconds
hive (iphone_analytics)> SHOW TABLES;
OK
bronze_customers
bronze_products
bronze_sales
bronze_stores
dim_customer
dim_date
dim_product
dim_store
fact_customers
fact_products
fact_sales
fact_stores
silver_customers
silver_products
silver_sales
silver_stores
Time taken: 0.063 seconds, Fetched: 16 row(s)
hive (iphone_analytics)> █
```

Hive Table Schema Validation

Screenshot must show:

DESCRIBE FORMATTED silver_sales;

```
hive (iphone_analytics)> DESCRIBE FORMATTED silver_sales;
OK
# col_name      data_type      comment
sale_id         int
product_id      int
customer_id     int
store_id        int
quantity        int

# Partition Information
# col_name      data_type      comment
sale_date       date

# Detailed Table Information
Database:        iphone_analytics
OwnerType:       USER
Owner:           takeo
CreateTime:      Fri Jun 26 08:24:58 IST 2026
LastAccessTime:  UNKNOWN
Retention:       0
Location:        hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_sales
Table Type:      MANAGED_TABLE
Table Parameters:
  numFiles              4
  numPartitions         4
  numRows               0
  rawDataSize           0
  spark.sql.create.version 3.3.2
  spark.sql.partitionProvider catalog
  spark.sql.sources.provider parquet
  spark.sql.sources.schema {\"type\":\"struct\",\"fields\": [{\"name\":\"sale_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"product_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"customer_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"store_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"quantity\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}]}
  spark.sql.sources.schema.numPartCols 1
  spark.sql.sources.schema.partCol.0   sale_date
  totalSize              5711
  transient_lastDdlTime 1782442499

# Storage Information
SerDe Library:      org.apache.hadoop.hive ql.io.parquet.serde.ParquetHiveSerDe
InputFormat:        org.apache.hadoop.hive ql.io.parquet.MapredParquetInputFormat
OutputFormat:        org.apache.hadoop.hive ql.io.parquet.MapredParquetOutputFormat
Compressed:         No
Num Buckets:        -1
Bucket Columns:     []
Sort Columns:       []
Storage Desc Params:
  path                hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_sales
  serialization.format 1
```

DESCRIBE FORMATTED fact_sales;

```
hive (iphone_analytics)> DESCRIBE FORMATTED fact_sales;
OK
# col_name      data_type      comment
sale_id         int
customer_id     int
product_id      int
store_id        int
quantity        int
total_amount     double

# Partition Information
# col_name      data_type      comment
date_key        date

# Detailed Table Information
Database:        iphone_analytics
OwnerType:       USER
Owner:           takeo
CreateTime:      Fri Jun 26 08:25:06 IST 2026
LastAccessTime:  UNKNOWN
Retention:       0
Location:        hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_sales
Table Type:      MANAGED_TABLE
Table Parameters:
  numFiles              4
  numPartitions         4
  numRows               0
  rawDataSize           0
  spark.sql.create.version 3.3.2
  spark.sql.partitionProvider catalog
  spark.sql.sources.provider parquet
  spark.sql.sources.schema {\"type\":\"struct\",\"fields\": [{\"name\":\"sale_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"customer_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"product_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"store_id\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"quantity\",\"type\":\"integer\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"total_amount\",\"type\":\"double\",\"nullable\":true,\"metadata\":{}}, {\"name\":\"date_key\",\"type\":\"date\",\"nullable\":true,\"metadata\":{}}]}
  spark.sql.sources.schema.numPartCols 1
  spark.sql.sources.schema.partCol.0   date_key
  totalSize              6811
  transient_lastDdlTime 1782442506

# Storage Information
SerDe Library:      org.apache.hadoop.hive ql.io.parquet.serde.ParquetHiveSerDe
InputFormat:        org.apache.hadoop.hive ql.io.parquet.MapredParquetInputFormat
OutputFormat:        org.apache.hadoop.hive ql.io.parquet.MapredParquetOutputFormat
Compressed:         No
Num Buckets:        -1
Bucket Columns:     []
Sort Columns:       []
Storage Desc Params:
  path                hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_sales
  serialization.format 1
```

Must clearly show:

- Storage format = Parquet
- Partition column (sale_date / date_key)
- Table location in HDFS

HDFS Directory Structure (Medallion Proof)

Screenshot must show:

```
hadoop fs -ls /warehouse/tablespace/managed/hive/
```

Evidence must include:

```
bronze_sales/
silver_sales/
fact_sales/
dim_product/
```

```
takeo@11b82bf4c789:~/startApps$ hadoop fs -ls hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db
2026-06-26 08:29:23,533 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 16 items
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/bronze_customers
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/bronze_products
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/bronze_sales
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/bronze_stores
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:13 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/dim_customer
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:13 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/dim_date
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:13 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/dim_product
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:13 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/dim_store
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_customers
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_products
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_sales
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_stores
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_customers
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_products
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:24 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_sales
drwxr-xr-x - takeo supergroup 0 2026-06-26 08:25 hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_stores
takeo@11b82bf4c789:~/startApps$
```

Partition Verification Screenshot

Screenshot must show:

```
SHOW PARTITIONS silver_sales;
```

```
hive (iphone_analytics)> SHOW PARTITIONS silver_sales;  
OK  
sale_date=2023-01-10  
sale_date=2023-01-12  
sale_date=2023-01-15  
sale_date=2023-01-20  
Time taken: 0.238 seconds, Fetched: 4 row(s)
```

SHOW PARTITIONS fact_sales;

```
hive (iphone_analytics)> SHOW PARTITIONS fact_sales;  
OK  
date_key=2023-01-10  
date_key=2023-01-12  
date_key=2023-01-15  
date_key=2023-01-20  
Time taken: 0.168 seconds, Fetched: 4 row(s)
```

Confirms:

- Partitioned by date
- Data is actually written correctly

Hive SQL Output (Business Queries)

Students must run at least 2 analytics queries.

Example:

```
SELECT product_name, SUM(total_amount)  
FROM fact_sales f  
JOIN fact_products p  
ON f.product_id = p.product_id  
GROUP BY product_name;
```

Screenshot must show:

- Query
- Output rows
- Column names

```

hive (iphone_analytics)> SELECT product_name, SUM(total_amount)
> FROM fact_sales f
> JOIN fact_products p
> ON f.product_id = p.product_id
> GROUP BY product_name;
Query ID = takeo_20260626083252_01809c90-17e2-4c1d-83d9-0ac6e4170e32
Total jobs = 1
SLF4J: Found binding in [jar:file:/home/takeo/bootcamp/installs/hive/apache-hive-3.1.3-bin/lib/log4j-slf4j-impl-2.17.1.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/takeo/bootcamp/installs/hadoop-3.3.3/share/hadoop/common/lib/slf4j-reloadj-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
2026-06-26 08:32:55 Starting to launch local task to process map join; maxinum memory = 239075328
2026-06-26 08:32:56 Uploaded 1 file to: file:/tmp/takeo/9b20e5ff-042f-4077-0635-e0a80081919c/hive_2026-06-26_08-32-52_302_3598954247238911017-1/-Local-10005/HashTable-Stage-2/MapJoin-mapfile21-..hashtable (345 bytes)
Execution completed successfully
MapredLocal task succeeded
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1782442237503_0003, Tracking URL = http://11b82bf4c789:8088/proxy/application_1782442237503_0003/
Kill Command = /home/takeo/installs/hadoop/bin/mapred job -kill job_1782442237503_0003
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2026-06-26 08:33:05,452 Stage-2 map = 0%, reduce = 0%
2026-06-26 08:33:09,665 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 2.73 sec
2026-06-26 08:33:13,763 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.89 sec
MapReduce Total cumulative CPU time: 3 seconds 890 msec
Ended Job = job_1782442237503_0003
MapReduce Jobs Launched:
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.89 sec HDFS Read: 33744 HDFS Write: 171 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 890 msec
OK
AlrPods 400.0
iphone 13 1800.0
iphone 14 2200.0
Time taken: 22.482 seconds, Fetched: 3 row(s)
hive (iphone_analytics)>

```

Row Count Validation

Screenshot must show:

SELECT COUNT(*) FROM bronze_sales;

```

hive (iphone_analytics)> SELECT COUNT(*) FROM bronze_sales;
Query ID = takeo_20260626083336_6a09d06a-2d4f-48d2-afed-34f93ab947f3
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1782442237503_0004, Tracking URL = http://11b82bf4c789:8088/proxy/application_1782442237503_0004/
Kill Command = /home/takeo/installs/hadoop/bin/mapred job -kill job_1782442237503_0004
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-06-26 08:33:43,259 Stage-1 map = 0%, reduce = 0%
2026-06-26 08:33:47,420 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 1.47 sec
2026-06-26 08:33:51,528 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 2.49 sec
MapReduce Total cumulative CPU time: 2 seconds 490 msec
Ended Job = job_1782442237503_0004
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 2.49 sec HDFS Read: 16682 HDFS Write: 101 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 490 msec
OK
4
Time taken: 16.157 seconds, Fetched: 1 row(s)

```

SELECT COUNT(*) FROM silver_sales;

```
hive (iphone_analytics)> SELECT COUNT(*) FROM silver_sales;
Query ID = takeo_20260626083352_826d6529-dd39-4b0a-9fb5-e791ea8221dd
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1782442237503_0005, Tracking URL = http://11b82bf4c789:8088/proxy/application_1782442237503_0005/
Kill Command = /home/takeo/installs/hadoop/bin/mapred job -kill job_1782442237503_0005
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-06-26 08:34:03,014 Stage-1 map = 0%, reduce = 0%
2026-06-26 08:34:07,197 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 2.32 sec
2026-06-26 08:34:10,260 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 3.54 sec
MapReduce Total cumulative CPU time: 3 seconds 540 msec
Ended Job = job_1782442237503_0005
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 3.54 sec HDFS Read: 25396 HDFS Write: 101 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 540 msec
OK
4
Time taken: 18.703 seconds, Fetched: 1 row(s)
```

SELECT COUNT(*) FROM fact_sales;

```
hive (iphone_analytics)> SELECT COUNT(*) FROM fact_sales;
Query ID = takeo_20260626083411_4beb896c-12c9-4f1d-95da-980ee59bfa33
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1782442237503_0006, Tracking URL = http://11b82bf4c789:8088/proxy/application_1782442237503_0006/
Kill Command = /home/takeo/installs/hadoop/bin/mapred job -kill job_1782442237503_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-06-26 08:34:20,170 Stage-1 map = 0%, reduce = 0%
2026-06-26 08:34:24,268 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 2.07 sec
2026-06-26 08:34:28,371 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 3.4 sec
MapReduce Total cumulative CPU time: 3 seconds 400 msec
Ended Job = job_1782442237503_0006
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 3.4 sec HDFS Read: 27555 HDFS Write: 101 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 400 msec
OK
4
Time taken: 18.073 seconds, Fetched: 1 row(s)
```

This proves:

- No silent data loss
- Pipeline integrity

Data Sampling Proof

Screenshot must show:

```
SELECT * FROM silver_sales LIMIT 10;
```

```
SELECT * FROM fact_sales LIMIT 10;
```

```
hive (iphone_analytics)> SELECT * FROM silver_sales LIMIT 10;
OK
1      101      1      10      1      2023-01-10
2      102      2      20      2      2023-01-12
3      101      3      20      1      2023-01-15
4      103      1      10      2      2023-01-20
Time taken: 1.056 seconds, Fetched: 4 row(s)
hive (iphone_analytics)> SELECT * FROM fact_sales LIMIT 10;
OK
1      1      101      10      1      900.0  2023-01-10
2      2      102      20      2      2200.0 2023-01-12
3      3      101      20      1      900.0  2023-01-15
4      1      103      10      2      400.0  2023-01-20
Time taken: 0.213 seconds, Fetched: 4 row(s)
```

Code Sharing Rules (Very Important)

Allowed:

- Shared GitHub repo for:
 - CSV readers
 - Schema definitions

Common Spark utilities